

1) Computer vision helps computer see, understand, and analyze images and videos like humans.

a) Main objectives of computer vision:

- To recognize objects
- To understand images
- To make decisions based on visual data.

b) Two real world applications:

- Face recognition
- Self-driving cars.

2) Computer vision can detect objects, recognize faces, and analyze images automatically.

a) Define feature extraction:

It detects, recognizes, and analyzes images

a) Feature extraction: Finding important image features

b) It helps identify objects accurately.

3) How are different levels of image processing used?

They improve, analyze, and understand images

a) Low level:

Enhances image quality by reducing noise and improving brightness.

- b) i) Mid-level: Detects features and segment objects
ii) High-level: Recognizes and understands objects

4) How is a digital image represented?

A digital image is made up of many tiny pixels arranged in rows and columns.

a) Pixel:

A pixel is the smallest unit of a digital image.

b) Resolution:

Higher resolution gives clearer and more detailed images.

5) How does image sensing and acquisition work?

It captures a real-world scene and converts it into a digital image.

a) Image Sensor:

Converts light into electrical signal.

b) Devices:

- Digital camera
- Smartphone camera.

6) How do sampling and quantization contribute to a digital image?

They convert a continuous image into digital form.

a) Sampling:

Divides the image into pixels.

b) Quantization:

Assigns ^{gr} brightness or color values to each point Pixel.

7) It explains how a camera captures an image using light and a lens.

a) Perspective projection:

converts a 3D scene into a 2D image.

b) illumination:

provides the light needed to capture the image.

8) How are image processing and computer vision related
Image processing improves images, while computer vision understands their content.

a) Image processing:

It enhances or modifies digital images.

b) Differences:

- Image processing improves images
- computer vision recognizes and interprets image.

9) How is computer vision used in Real-world Application?

It helps solve problems by analyzing visual information automatically.

a) Healthcare:

Detects diseases from X-rays and MRI scans

b) Autonomous vehicles:

Detects roads, vehicle, and traffic signal.

10. How do digital image fundamentals support computer vision.

They provide the basic information needed for image analysis

a) pixels

pixels store the image information.

b) Sampling and Resolution:

Highest sampling and resolution improve image quality and analysis accuracy.